Operating System (BCA 4th Semester) - Full 2 Mark Answers (Q1-Q31)

1. Define an Operating System?

An Operating System (OS) is system software that acts as an interface between the user and computer hardware and manages all computer resources efficiently.

2. What is a Process?

A process is a program that is currently being executed. It includes the program code, data, and resources required for execution.

3. What are the drawbacks of Batch Processing System?

No direct interaction between user and computer, long waiting time for job completion, and difficulty in identifying and correcting errors.

4. What is Time Sharing System and Multiprogramming System?

Time Sharing System shares CPU time among multiple users or tasks. Multiprogramming System keeps multiple programs in memory simultaneously and switches among them to maximize CPU utilization.

5. What is Real-Time System?

A Real-Time System is an operating system that provides immediate processing and response within a specified time limit.

6. Write the two types of Real-Time System.

Hard Real-Time System and Soft Real-Time System.

7. Define Distributed System.

A Distributed System is a collection of independent computers connected through a network that work together as a single system.

8. What is Multiprocessor System?

A Multiprocessor System is a computer system that contains two or more CPUs sharing the same memory and resources.

9. Give two advantages of Multiprocessor System.

Increased processing speed and performance; improved reliability and fault tolerance.

10. Write the two types of Multiprocessor System.

Asymmetric Multiprocessing (AMP) and Symmetric Multiprocessing (SMP).

11. Define Symmetric Multiprocessor System.

A Symmetric Multiprocessor System is a multiprocessor system in which all processors are equal and share the same memory and operating system. Each processor can perform any task.

12. Define Asymmetric Multiprocessor System.

An Asymmetric Multiprocessor System is a system where one processor acts as the master and controls the other processors by assigning them tasks.

13. List any two functions of Process Management Component of the OS.

Creating and deleting processes; scheduling processes for CPU execution.

14. List any two functions of Memory Management Component of the OS.

Allocating and deallocating memory to processes; keeping track of memory usage.

15. List any two functions of File Management Component of the OS.

Creating and deleting files; managing file storage and access permissions.

16. List any two functions of Input/Output System Management Component of the OS.

Managing I/O devices and drivers; coordinating data transfer between devices and memory.

17. List any two functions of Networking Management Component of the OS.

Managing network communication between computers; providing network security and access control.

18. Write any four operating system services.

Program execution, I/O operations, file system manipulation, and communication.

19. Give any four attributes of File.

File Name, File Type, File Size, and Creation Date and Time.

20. Give any four File Types.

Text File, Executable File, Source Code File, and Object File.

21. Give any four File Operations.

Create, Open, Read, and Write.

22. Give any four Directory Operations.

Create a directory, Delete a directory, Search for a file, and Rename a directory.

23. List any two file access methods.

Sequential Access and Direct (Random) Access.

24. Define sequential access with respect to files.

In Sequential Access, file records are accessed one after another in order, starting from the beginning of the file.

25. Define random access with respect to files.

In Random (Direct) Access, any record in a file can be accessed directly without reading the preceding records.

26. List four types of file accesses.

Sequential Access, Direct Access, Indexed Access, and Indexed Sequential Access.

27. List any four allocation methods.

Contiguous Allocation, Linked Allocation, Indexed Allocation, and Extent-Based Allocation.

28. Define contiguous allocation.

Contiguous Allocation is a file allocation method in which all blocks of a file are stored in consecutive memory locations on the disk.

29. What is linked allocation?

Linked Allocation is a file allocation method in which file blocks are linked together using pointers, and the blocks may be stored anywhere on the disk.

30. What is indexed allocation?

Indexed Allocation is a file allocation method in which each file has an index block that contains the addresses of all the file's data blocks. This allows direct access to file blocks.

31. List four Free Space Management.

Bit Vector (Bitmap) Method, Linked List Method, Grouping Method, and Counting Method.